Pascal's triangle

1

a $1, 5, 10, 10, 5, 1$

b 32

2 $1, 8, 28, 56, 70, 56, 28, 8, 1$

3 7th

4

a 7

b 9

5

a 5

b 4

6 $(x+m)^4 = 1x^4m^0 + 4x^3m^1 + 6x^2m^2 + 4x^1m^3 + 1x^0m^4$

Binomial theorem

7

a 6

b 14

c 10

d 1

8 Negative

9 4

10

a $256x^4$

b $81y^4$

11

a $560b^3$

b $126c^5d^4$

c $135u^2v^4$

d $\frac{55}{128}u^3$

e $\frac{280}{3y^2}$

f $2288x^3y^{20}$

g $-3876x^4y^{75}$

h $1\,451\,520x^8y^{12}$

i $216x^{-6}y^{-4}$

j 1792

12 $45x^2y^8, 10xy^9, y^{10}$

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13 **a** $n = 10$ **b** $45m^2k^8$

14 $p = 2$

15 **a** 10 **b** 4th term

16 $55u^2v^9, 55u^9v^2$

17 10 777 536

18

a $1 + 4x + 6x^2 + 4x^3 + x^4$

b $32 + 80b + 80b^2 + 40b^3 + 10b^4 + b^5$

c $p^5 + 5p^4q + 10p^3q^2 + 10p^2q^3 + 5pq^4 + q^5$

d $c^8 - 8c^7d + 28c^6d^2 - 56c^5d^3 + 70c^4d^4 - 56c^3d^5 + 28c^2d^6 - 8cd^7 + d^8$

e $u^6 + 24u^5v + 240u^4v^2 + 1280u^3v^3 + 3840u^2v^4 + 6144uv^5 + 4096v^6$

f $y^4 + 12y^3 + 54y^2 + 108y + 81$

g $(ka)^6 - 6(ka)^5 b + 15(ka)^4 b^2 - 20(ka)^3 b^3 + 15(ka)^2 b^4 - 6kab^5 + b^6$

h $y^3 - 15y^2 + 75y - 125$

i $y^3 - y^2 + \frac{y}{3} - \frac{1}{27}$

j $8 + 6y + \frac{3y^2}{2} + \frac{y^3}{8}$

k $27y^3 + 54ry^2 + 36r^2y + 8r^3$

l $a^3 - 3a + \frac{3}{a} - \frac{1}{a^3}$

m $u^6 + 9u^4v^2 + 27u^2v^4 + 27v^6$

n $1 + \frac{12}{y} + \frac{48}{y^2} + \frac{64}{y^3}$

19 **a** $405x^4$ **b** $120c^3d^7$ **c** $165u^3v^8$ **d** $3003u^5v^{10}$

20 $n = 16$

21 $\frac{21}{2}$

22 84

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23 9

24

a 0.943 **b** 1.037 **c** 0.890

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